Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/12/2026
PI Name:	Rothschild, Lynn Ph.D.		
Project Title:	Defining the Figures of Merit for an On-demand Astropharmacy		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 TRISH FOM BEES 2024
Start Date:	09/01/2024	**End Date:**	08/31/2025
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-FMBS0102		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	As NASA embarks on a new era in space exploration beyond low-Earth orbit (LEO), the need to provide effective pharmaceuticals in space must be addressed. The state-of-the-art is to bring drugs on board, but this approach has problems that are exacerbated by long-duration missions. First, the exact needs of the crew cannot be anticipated, resulting in the requirement for a drug inventory that may be excessive or inadequate. Second, drugs are not FDA rated for long-term stability. NASA has encouraged development of methods to enhance stability, but that does not address the first issue. Peptide/protein drugs pose the greatest challenge in terms of stability. Even with refrigeration, these drugs have the shortest shelf-life (months), as compared to small organic molecule drugs with shelf lives of years. Peptide/protein drugs, therefore, require a “production-on-demand” solution for multi-year, deep space missions. The Astropharmacy innovation was developed with a laser-like focus on the goal of providing “on-demand” production of peptide/protein-based pharmaceuticals for astronauts at scales that are suitable for a short course treatment for a medical condition that represents immediate threat to the health and well-being of an individual crewmember. The overall aim of		

	the project is to derive the Figures of Merit (FOM; mass, volume, power, astronaut time) for the two types of production systems we have developed for Astropharmacy, one based on traditional growth and peptide purification protocols used in the lab, and the other using a spaceflight hardware prototype. This will provide a snapshot of where the system and the prototypes are in terms of FOM, as well as suggest where they can be improved in future hardware/protocol development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The conducted analysis provided a snapshot of the current state of the system and the prototype and provided insights into how the FOM would change as the culture volume needed to produce doses of different drugs changes. The FOM analysis also helped identify areas of improvement for future hardware/prototype development. In particular, using custom heating/agitation designs can significantly reduce FOM associated with lab incubators that are not optimized for spaceflight. Additionally, implementing alternative ways to separate cell and debris from soluble lysate fraction that do not require centrifugation could greatly reduce the mass/volume/power requirements for the spaceflight hardware prototype. Lastly, implementation of automation in the spaceflight hardware prototype helped to greatly reduce crew time requirements, compared to the traditional laboratory procedures. Overall, these developments are critical steps towards implementation of Astropharmacy to address a critical need to provide effective pharmaceuticals for long-duration missions beyond low-Earth Orbit. On Earth, the ability to produce small quantities of on-demand pharmaceuticals with minimal power, mass, and storage demands should be of enormous benefit in field situations where time is of the essence or resupply impossible. Such situations include remote areas, including the Antarctic, and extended submarine deployments. This approach could make orphan drugs for terrestrial applications more affordable, enabling more personalized medical treatment.
Bibliography Type:	Description: (Last Updated: 05/12/2026)
NASA Technical Documents	Kunitskaya A, Snyder JE, Blum RC, Omrani D, Loftus DJ, Rothschild LJ. "Astropharmacy: in-Space Pharmaceutical Manufacturing for Deep Space Missions." Moffett Field, California: NASA Ames Research Center, 2025. NASA Document ID 20250009075. https://ntrs.nasa.gov/api/citations/20250009075/downloads/20250009075-IAC2025_manuscript_Kunitskaya.pdf , Sep-2025
Papers from Meeting Proceedings	Kunitskaya A, Snyder JE, Blum RC, Omrani D, Loftus DJ, Rothschild LJ. "Astropharmacy: in-space pharmaceutical manufacturing for deep space missions." 76th International Astronautical Congress (IAC), Sydney, Australia, Sep 29-Oct 3, 2025. 76th International Astronautical Congress (IAC), Sydney, Australia, Sep 29-Oct 3, 2025. IAC-25,A2,8,6,x100751. , Sep-2025